Sourav Selvaraj

Research interests: contact-rich control and learning for legged systems.

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Education

- **Worcester Polytechnic Institute** — Worcester, MA
  Master of Science in Robotics Engineering — *Aug 2025 – May 2027 (expected)*
  - **GPA:** 4.0/4.0
  - **Coursework:** Robot Control, Motion Planning, Legged Robotics, Machine Learning for Robotics
- **National Institute of Technology, Tiruchirappalli** — Tiruchirappalli, India
  B.Tech in Instrumentation and Control Engineering — *Dec 2021 – May 2025*
  - **CGPA:** 8.2/10.0
  - **Coursework:** Control Systems, Robotics, Signal Processing, Machine Learning, Embedded Systems

Manuscripts

[1] **S. Selvaraj** and M. Agheli, "SCRUB: Contact-Based Surface-Frame Estimation for Force–Motion Control with an Armless Quadruped." To be submitted to IEEE RA-L, 2026.
[2] O. Sullivan*, **S. Selvaraj***, and M. Agheli, "GaitNet: Where Do I Step?" To be submitted to L4DC 2027. *Equal contribution.

Research Experience

- **SCRUB: Armless Quadruped Body-Tool Sliding** [video] — ALMaS Research Group, WPI
  Graduate Researcher; Advisor: Dr. Mahdi Agheli — *Aug 2025 – Present*
  - Developed **SCRUB** for force-controlled tool sliding on a Unitree Go1 without a surface model.
  - Derived a friction-independent surface-normal estimator from tool forces.
  - Extended a **100 Hz** OCS2 centroidal MPC to plan the tool's contact, over a **1 kHz** whole-body controller.
  - Completed **5/5** wall and **5/5** column trials at 10 N, with **2.29°/2.48°** normal error and **2.97/2.06 N** force RMSE; fixed-normal and friction-compensated baselines completed **0/5** column trials.
- **GaitNet: Attention-Based RL Footstep Planning** [docs] — ALMaS Research Group, WPI
  Graduate Researcher; Advisor: Dr. Mahdi Agheli — *Nov 2025 – Present*
  - Implemented **GaitNet-T** with 4-head attention over candidate footholds and a motion-history encoder.
  - Built controller-specific **IDMs**: **0.89 cm vs 2.61 cm** error and **83% vs 55%** OCS2 success.
  - In simulation with camera terrain, GaitNet reached **71%** (5 seeds) at 40–70% voids vs **38%** for OCS2 trot.
  - Ran the frozen policy on a real Go1 with OCS2, stepping acyclically between 40–50 mm blocks.
- **TOTO: Autonomous Mobile Robot** — NIT Trichy, India
  Undergraduate Researcher; Advisor: Dr. D. Ezhilarasi — *Sep 2024 – May 2025*
  - Built an **ICP-SLAM** stack with an Iterated Kalman Filter fusing encoders and RPLiDAR.
  - Achieved ±2% velocity-tracking error and reduced point-cloud processing by **10%**.
- **Biomimetic Autonomous Underwater Vehicle** — IIT Madras, India
  Research Intern; Advisor: Dr. Asokan Thondiyath — *Mar 2023 – Sep 2023*
  - Designed a Jetson Orin interface PCB and CAN motor control for Maxon BLDC propulsion drives.

Honors & Awards

- **Dr. Glenn Yee Graduate Student Tuition Award** — WPI, Worcester
  Awarded to graduate students for academic excellence — *Fall 2026*
- **Winner, Smart India Hackathon** — Ministry of Education, India
  Awarded in a nationwide student innovation competition — *2023*

Selected Projects

Code and demonstration videos: `github.com/souravselvaraj`

- **Humanoid Ice Skating with RL** [video] *Jul 2026 – Present*
  - Trained a Unitree G1 in simulation to skate with PPO on passive 3 mm blades without reference motions.
  - Tracked 0.9 m/s with **0.09 m/s** error, **94%** single support, and **2–4 cm/s** lateral slip.
- **Humanoid Bimanual Lifting with Residual AMP–PPO** [code] *May 2026 – Jul 2026*
  - Collaborated on training a G1 to lift objects via residual AMP–PPO over a fixed locomotion policy.
  - Held objects within **3 cm** of the target in simulation.
  - Ran the lifting policy on the real G1 with a box.
- **Crazyflie SE(3) Control with L1 Adaptation** [code] *Oct 2025 – Dec 2025*
  - Implemented an SE(3) controller with L1 adaptation for a Crazyflie 2.1 in ROS 2.
  - Achieved **2.2 cm** position RMSE and rejected **94%** of force-step disturbances.

Additional: Flow matching (`code`); ORB-SLAM; visuomotor imitation (`code`); PDDL/OMPL.

Teaching Experience

- **RBE 521: Legged Robotics; Parallel Robotics** Worcester Polytechnic Institute
  Teaching Assistant & Laboratory Curriculum Developer *May 2026 – Aug 2026*
  - Served as RBE 521 TA with Dr. Mahdi Agheli and developed Unitree Go1/G1 lab modules.
  - Developed hands-on lab modules for Parallel Robotics with Dr. McCann.

Academic Service & Leadership

- **Reviewer:** *IEEE Transactions on Automation Science and Engineering (T-ASE)*, Jan–Feb 2026.
- **Team Lead, ISRO Robotics Challenge:** Led a 10-person autonomous-rover team, Aug 2024–May 2025.

Skills

Software: ROS 2, Isaac Lab, MuJoCo, Gazebo, OCS2, Pinocchio, PyTorch, Docker
Programming: C/C++, Python, MATLAB/Simulink
Robotics: MPC, Whole-Body Control, Trajectory Optimization, Reinforcement Learning, Imitation Learning
Hardware: Unitree Go1/G1, Franka Panda, Crazyflie 2.1, Jetson Orin, RealSense, F/T Sensors